

Candidate Name

Centre Number

Candidate Number



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

BUILDING TECHNOLOGY & DESIGN

6003/1

PAPER 1 Theory and Drawing.

3 hours

SPECIMEN PAPER

Additional materials:

Answer paper,
 Plain paper A4 (1 sheet),
 Scientific calculator.

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page and on **all** separate answer paper used.

Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B

Answer any **two** questions.

Write your answers on the separate answer paper provided.

At the end of the examination, fasten the separate answer paper securely to the question paper.

Section C

Compulsory question.

Write your answer on the separate answer paper provided.

At the end of the examination, fasten the separate drawing paper securely to the question paper.

INFORMATION FOR CANDIDATES

Marks are given in brackets [] at the end of each question or part question.

All dimensions are in millimetres unless otherwise stated.

FOR EXAMINER'S USE

Section A	
Section B	
Section C	
TOTAL	

This question paper consists of 10 printed pages and 2 blank pages.

Copyright: Zimbabwe School Examinations Council, Specimen Paper.

Section A (50 Marks)

Answer **all** questions in the spaces provided.

For
Examiner's
Use

- 1 (a)** Illustrate **four** ways of terminating dimension lines of drawings.

_____ [1]

_____ [1]

_____ [1]

_____ [1]

- (b)** Explain the following terms as they are used in site surveying:

- (i)** datum,

_____ [2]

- (ii)** back-sight,

_____ [2]

- (iii)** rise.

_____ [2]

- 2 (a) Identify any **five** causes of ill-health associated with the building industry.

[5]

- (b) List any **five** effects of accidents at a building site.

[5]

- 3 (a) (i) State any **two** provisions of site security.

[2]

- (ii) Justify the need for site security control on a construction site.

[2]

(b) Explain the following types of waste disposal methods:

(i) recycling,

[2]

(ii) composting and

[2]

(iii) reclaiming.

[2]

- 4 (a) Explain the relationship between retention fee and defects liability period.

[4]

- (b) Describe the procedure of patenting.

[6]

**For
Examiner's
Use**

- 5 (a) Differentiate *renovation* and *replacement*.

[2]

- (b) State the importance of the following legislations as used in building control:

- (i) Planning permission,

[2]

- (ii) Building regulations,

[2]

- (iii) Health and safety control.

[2]

- (c) Why do Architects sometimes nominate subcontractors?

[2]

**For
Examiner's
Use**

SECTION B (30 Marks)

*Answer any **two** questions from this section. Each question carries **15** marks.*

- 6** **(a)** Define *bulking of sand*. [2]
- (b)** Justify the need for testing building materials before use. [2]
- (c)** Identify **three** properties of concrete in its:
- (i)** wet state, [3]
- (ii)** hardened state. [3]
- (d)** Outline the procedure of carrying out a cube test on concrete. [5]
- 7** Outline the open tendering procedure for choosing a building contractor. [15]
- 8** Discuss any **five** characteristics of Zimbabwean architecture. [15]
- 9** **(a)** With reference to structural design work of built structures, write brief notes on footing and support your answer with a neatly labelled diagram. [5]
- (b)** Compare using any **five** points, concrete and steel as structural materials of the built environment. [10]
- 10** With the aid of a diagram explain the production of biogas in the biogas digester. [15]

Section C

Compulsory Question

- 11 (a) **Fig. 11** shows a sketch of an incomplete site layout with a proposed building structure.

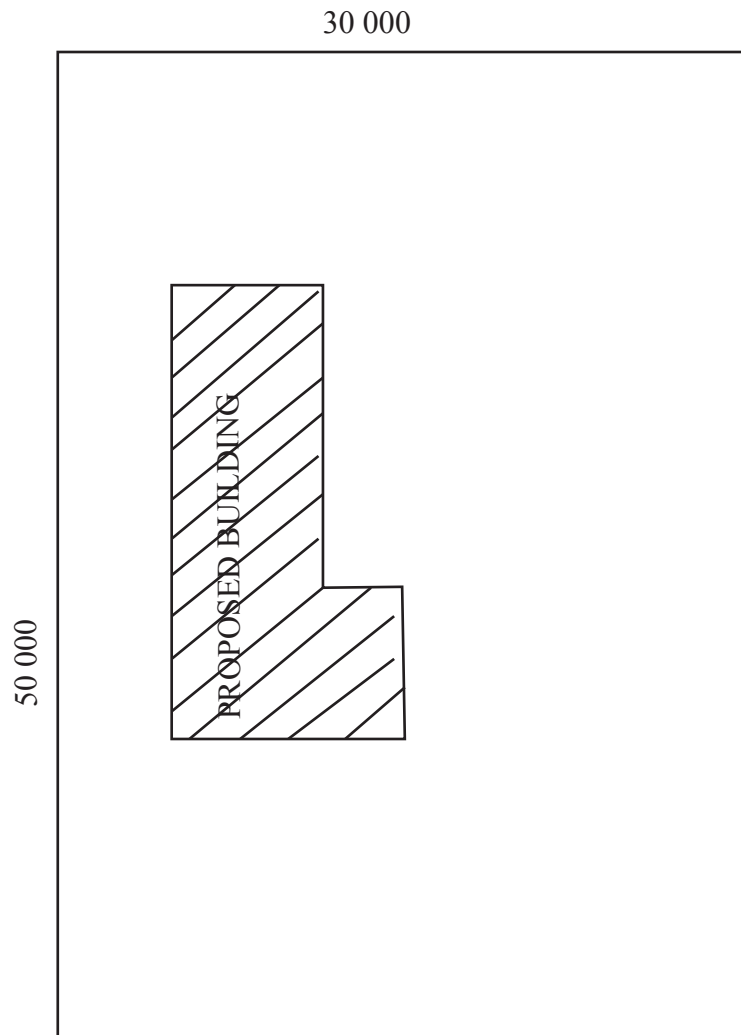


Fig. 11

Design and draw the detailed site layout showing suitable arrangements of site provisions to promote the following:

security, [3]

services, [3]

10

hutments, [2]

welfare facilities, [3]

material storage, [3]

plant and equipment, [3]

informative signs. [2]

NB Give the overall dimensions. [1]